Aim:- Prepare a detailed Software Requirement Specification (SRS) document for the Hospital
Management System as per IEEE standard.

Project Title Hospital Management System

Software Requirement Specification (SRS)

1. Introduction

1.1 Purpose

This document describes the **Software Requirements Specification (SRS)** for a **Hospital Management System**. The system will manage patient records, doctor information, appointments, billing, laboratory reports, pharmacy services, and hospital administration through a centralized web-based application.

1.2 Scope

The **Hospital Management System** aims to provide an online platform for hospitals to efficiently manage patient information, doctor schedules, appointments, medical records, laboratory reports, pharmacy services, billing, and administrative tasks. It helps improve hospital efficiency, reduce paperwork, and provide better healthcare services.

1.3 Definitions, Acronyms, Abbreviations

- **SRS:** Software Requirements Specification
- **HMS:** Hospital Management System
- **Admin:** Hospital administrator who manages users and system data
- **DB:** Database
- **OPD:** Outpatient Department
- **IPD:** Inpatient Department
- **EMR:** Electronic Medical Record

1.4 References

- IEEE 830-1998 Software Requirements Specification Standard
- Hospital Management Policies
- Healthcare Management Guidelines
- Sample Hospital Management Systems

1.5 Overview

This SRS describes the overall description, functional requirements, non-functional requirements, interfaces, constraints, assumptions, and supporting information for the Hospital Management System.

2. Overall Description

2.1 Product Perspective

The Hospital Management System is a stand-alone web application. In the future, it can be integrated with online payment gateways, telemedicine services, health insurance systems, and electronic health record (EHR) platforms.

2.2 Product Functions

- Register new patients
- Manage patient records
- Book and manage appointments
- Manage doctor schedules
- Generate prescriptions
- Maintain laboratory reports
- Manage pharmacy inventory
- Generate bills and payment receipts
- Generate hospital reports
- Enable admin to manage users and hospital data

2.3 User Characteristics

- **Patients:** Register, book appointments, view reports, and download prescriptions.
- **Doctors:** View appointments, update diagnosis, prescribe medicines, and access patient records.
- **Receptionists:** Register patients, schedule appointments, and generate bills.
- **Pharmacists:** Manage medicines and issue pharmacy bills.
- **Laboratory Staff:** Upload laboratory reports and manage test requests.
- **Admin:** Manage users, doctors, reports, and hospital information.

2.4 Constraints

- Must follow healthcare data security standards.
- Must provide secure login for all users.
- Must be compatible with major web browsers.
- Must maintain patient data confidentiality.
- Must comply with local healthcare regulations.

2.5 Assumptions & Dependencies

- Reliable internet connection is available.
- Hospital server is properly configured.
- Doctors and staff provide accurate information.
- Database server is always available.
- Users have valid login credentials.

3. Specific Requirements

3.1 Detailed Functional Requirements (Working Criteria)

- **FR-01:** The patient shall be able to register online.
- **FR-02:** The patient shall be able to book, update, or cancel appointments.
- **FR-03:** The doctor shall be able to view scheduled appointments.
- **FR-04:** The doctor shall be able to update patient diagnosis and prescriptions.
- **FR-05:** The receptionist shall be able to register patients.
- **FR-06:** The system shall securely store patient records in the database.
- **FR-07:** The laboratory staff shall be able to upload test reports.
- **FR-08:** The pharmacist shall be able to manage medicine inventory.
- **FR-09:** The system shall automatically generate hospital bills.
- **FR-10:** The administrator shall be able to log in securely.
- **FR-11:** The administrator shall be able to add, edit, and delete doctor, staff, and patient information.
- **FR-12:** The system shall generate daily, monthly, and yearly hospital reports.

3.2 Non-Functional Requirements

Performance

- Pages should load within **3 seconds**.
- The system should support multiple users simultaneously.

Storage

- The system must securely store patient records, prescriptions, laboratory reports, billing information, and medical history.

Security

- Secure login for all users.
- Password encryption.
- Role-based access control.
- Patient information must remain confidential.

Usability

- User-friendly interface for patients and hospital staff.
- Easy navigation and responsive design.

Interoperability

- Compatible with Chrome, Firefox, Edge, and Safari browsers.

Flexibility

- New departments, doctors, and services can be added in the future without major system changes.

Disaster Recovery

- Automatic database backup.
- Recovery mechanism in case of server failure.

3.3 Interfaces

User Interface

- Patient Registration Page
- Login Page
- Appointment Booking Page
- Doctor Dashboard
- Admin Dashboard
- Billing Page

- Pharmacy Management Page
- Laboratory Report Page
- Patient Medical Record Page

Hardware Interface

- Runs on standard computers, laptops, printers, barcode scanners, and hospital servers.

Software Interface

- Compatible with Apache Web Server
- MySQL Database
- PHP/Java/Python Backend
- Windows/Linux Operating System

Communication Interface

- Online appointment booking
- Email notifications
- SMS appointment reminders
- Contact and enquiry forms

4. Appendices

- Sample patient registration form
- Sample appointment booking form
- Sample medical report
- Sample hospital bill format
- Sample prescription format
- Hospital department list
- Glossary of hospital management terms